

Scattering by a central Potential

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$$d^3r = r^2 dr \underbrace{d\phi \sin\theta d\theta}_{d\Omega}$$

INCIDENT ENERGY $E = \frac{m}{2} v_0^2$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

ANGULAR MOMENTUM $l = S m v_0 = S \sqrt{2mE}$

$$S = \frac{l}{\sqrt{2mE}}$$

APSIDAL POINT

SCATTERED RING

SOLID ANGLE: $d\Omega = 2\pi \sin\theta d\theta$

$$S(\theta, E)$$

Azimuthal angle ϕ irrelevant by symmetry

PARTICLES EMERGING WITH SCATTERING ANGLE BETWEEN θ AND $\theta + d\theta$

DIFFERENTIAL CROSS SECTION

INCIDENT RING AREA: $2\pi S ds$

PARTICLES INCIDENT WITH IMPACT PARAMETER BETWEEN S AND $S + dS$

$S \leftrightarrow \theta$

INCIDENT INTENSITY

AREA: $2\pi s ds$

BETWEEN s AND $s+ds$



ANGLE BETWEEN θ AND $\theta+d\theta$



DIFFERENTIAL CROSS SECTION

INCIDENT INTENSITY

$$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$$

$$\sigma(\Omega) d\Omega = \frac{\text{\# PART SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$$

$$I (2\pi s) |ds| = \text{\# part going through ring between } s \text{ and } s+ds = \text{\# part emerging with angle between } \theta \text{ and } \theta+d\theta = I \sigma(\theta) 2\pi \sin\theta |d\theta|$$

THEREFORE:

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|$$

DEFINE ALSO THE TOTAL CROSS SECTION:

$$\sigma_T = \int d\Omega \sigma(\Omega)$$

$$\sin \theta |d\theta|$$

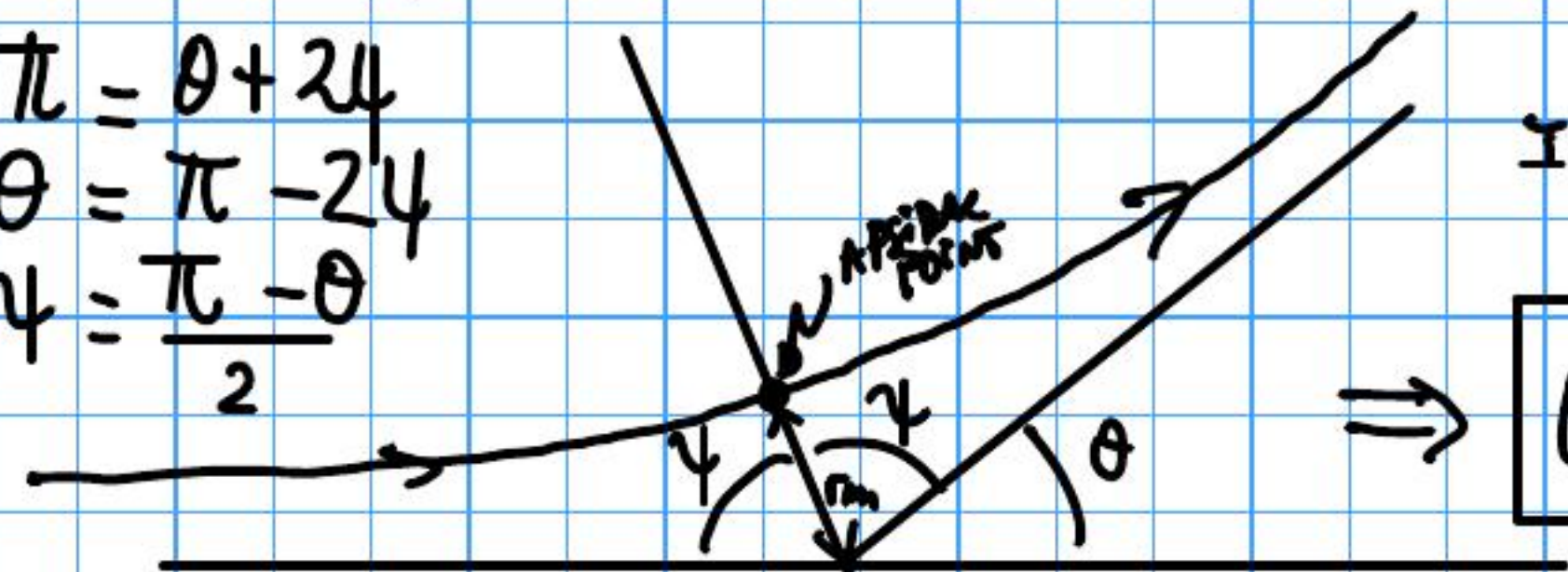
CROSS SECTION: σ

REPULSIVE CASE:

$$\pi = \theta + 2\psi$$

$$\theta = \pi - 2\psi$$

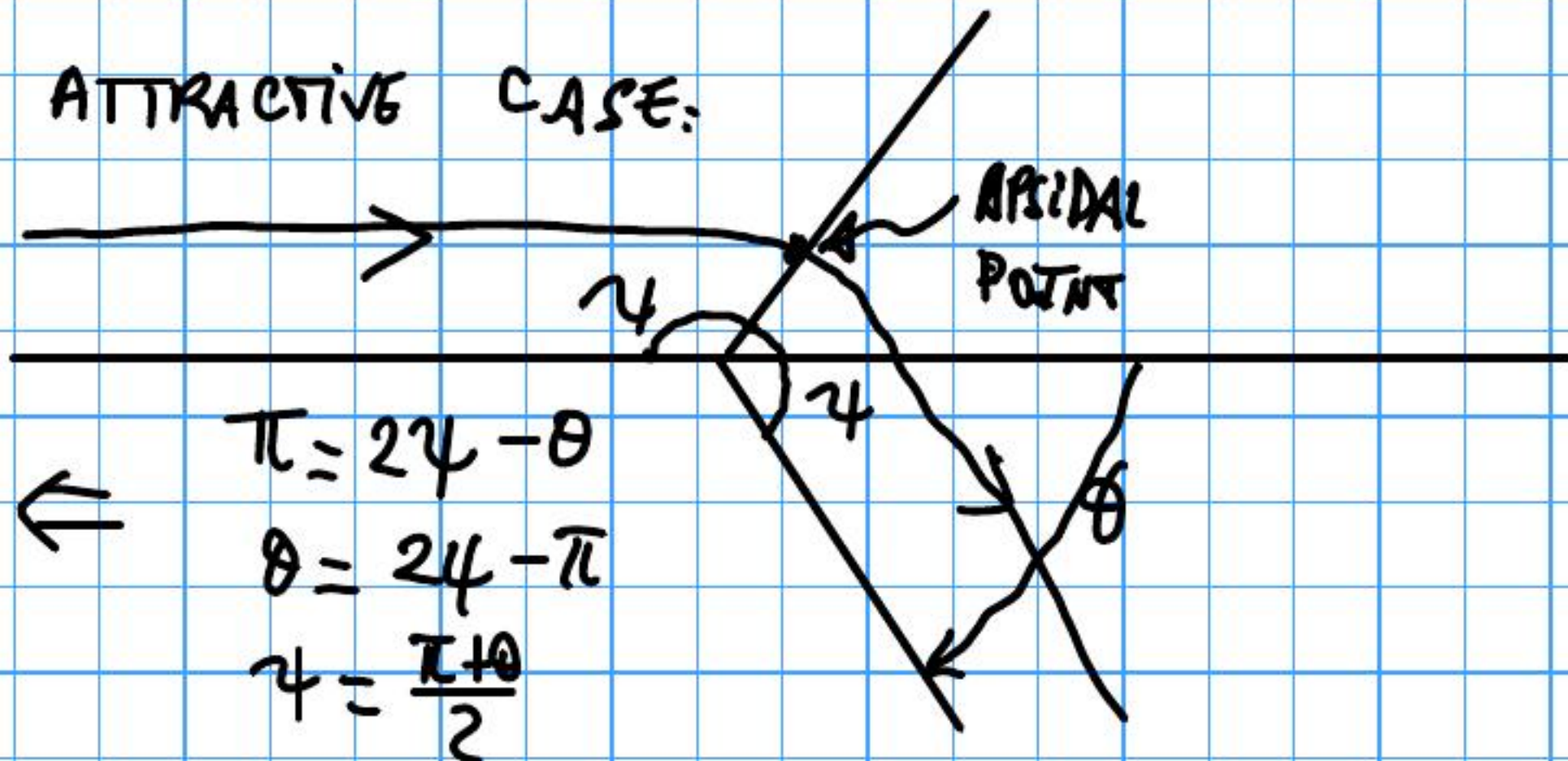
$$\psi = \frac{\pi - \theta}{2}$$



IN ALL
CASES:

$$\theta = |\pi - 2\psi|$$

ATTRACTIVE CASE:



$$\pi = 2\psi - \theta$$

$$\theta = 2\psi - \pi$$

$$\psi = \frac{\pi + \theta}{2}$$

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2} \quad \dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2} \quad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

$$l = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l}{mr^2} \frac{dr}{\dot{r}} = \frac{l}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}} = \frac{l}{m} \sqrt{\frac{m}{2E}} \frac{r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \frac{l^2}{2mE} \frac{1}{r^2} \right]^{1/2}}$$

$$\gamma = \int^4 d\gamma = \int^{\pi} d\theta = \int^{\infty} \frac{s r^{-2} dr}{\left[1 - V(r) - s^2 \right]^{1/2}} \quad = \frac{l}{\sqrt{2mE}} = s \quad \left[= s^2 \right]$$

$$\psi = \int_0^{\theta} d\psi = \int_{\theta_m}^{\theta} d\theta = \int_{r_m}^r \frac{s \, dr}{\left[1 - V(r) - \frac{s^2}{r^2}\right]^{1/2}} \Rightarrow \boxed{\theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s \, du}{\left[1 - \frac{V(u)}{E} - s^2 u^2\right]^{1/2}} \right|}$$

$= \frac{\sqrt{2mE}}{\hbar} s$
 $\int = s^2$

$u = 1/r \quad du = -r^{-2} dr \quad u_m = 1/r_m$

From here, compute $\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|}$ and use it to get $\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right|$

UP TO HERE, RESULTS ARE VALID FOR ANY CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

$$E > 0$$

hyperbolic

$$E = 0$$

parabolic

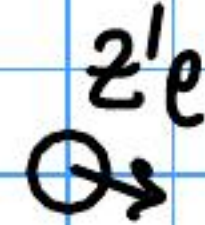
$$E < 0$$

elliptic

ONLY FOR $V = -\frac{k}{r}$!!!

FORCE

$$F = \frac{Z Z' e^2}{r^2}$$



$$f = -\frac{k}{r^2} \quad V = -\frac{k}{r}$$

$$k = -Z Z' e^2$$

Scattering: $E > 0$

$$\frac{1}{r} = G \left[1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}') \right]$$

$$G = \frac{m k}{l^2} = -\frac{m}{l^2} Z Z' e^2$$

$$\text{Eccentricity } \epsilon = \left[1 + \frac{2 E l^2}{m k^2} \right]^{\frac{1}{2}} = \left[1 + \frac{2 E l^2}{m (Z Z' e^2)^2} \right]^{\frac{1}{2}} = \left[1 + \left(\frac{2 s E}{Z Z' e^2} \right)^2 \right]^{\frac{1}{2}} > 1$$

$$l = s m v_0 = s \sqrt{2 m E} \rightarrow \frac{2 l^2 E}{m} = \frac{s^2 2^2 m E^2}{m} = (2 s E)^2$$

2 SITUATIONS: i) REPULSIVE CASE $zz' > 0 \Rightarrow k = -zz'e^2 < 0$

ii) ATTRACTIVE CASE $zz' < 0 \Rightarrow k = -zz'e^2 > 0$ (ALSO GRAVITY $k = Gm_1m_2$)

i) REPULSIVE CASE: $k < 0$ $C = \frac{mk}{l^2} < 0$

$$\frac{1}{r} = C[1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 \Leftrightarrow 1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\epsilon}$$

We need the cosine to be negative and larger in magnitude than $1/\epsilon$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta}' - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = C[1 + \epsilon] < 0$
UNPHYSICAL

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \underbrace{\cos \tilde{\theta} \cos \pi}_{=1} + \underbrace{\sin \tilde{\theta} \sin \pi}_0 = -\cos \tilde{\theta}$$

$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right) [-1 - \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{mzz'e^2}{l^2} [\epsilon \cos \tilde{\theta} - 1]$$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

$$\text{MINIMUM } r: \frac{1}{r_m} = \frac{mzz'e^2}{l^2} [\epsilon - 1]$$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

[IN THE REPULSIVE CASE]
 $\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2}$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

ii) ATTRACTIVE CASE: $k > 0$ $C = \frac{km}{l^2} > 0$

TAKE $\tilde{\theta}' = 0$ $\tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C[1 + \epsilon \cos(0)] = C[1 + \epsilon]$

MAX AT $\tilde{\theta} = 0$

MINIMUM r : $\frac{1}{r_m} = C[1 + \epsilon]$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

[IN THE ATTRACTIVE CASE]
 $\theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2}$

$$0 = 1 + \epsilon \cos \psi \Leftrightarrow -\frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}$$

We now combine both cases:

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{Z Z' e^2} \right)^2 \Rightarrow \boxed{s = s(\theta, E) = \left| \frac{Z Z' e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} \left| \frac{Z Z' e^2}{2E} \right| \frac{d}{d\theta} \left(\cot \frac{\theta}{2} \right) = \left| \frac{Z Z' e^2}{2E} \right| \left(\frac{1}{2} \left(-\frac{\sin \frac{\theta}{2}}{\cos^2 \frac{\theta}{2}} \right) - \frac{1}{2} \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{Z Z' e^2}{2E} \right| \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{2Z'e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \theta/2}{\sin \theta/2} \right) = \left| \frac{2Z'e^2}{2E} \right| \left(\frac{1}{2} \left(\frac{-\sin \theta/2}{\sin^2 \theta/2} \right) - \frac{1}{2} \frac{\cos^2 \theta/2}{\sin^2 \theta/2} \right) = - \left| \frac{2Z'e^2}{4E} \right| \frac{1}{\sin^2 \theta/2}$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{2Z'e^2}{2E} \right)^2 \frac{1}{2 \sin \theta/2 \cos \theta/2} \frac{1}{\sin^2 \theta/2}$$

$$\sigma(\theta) = \frac{1}{4} \left(\frac{2Z'e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)$$

Rutherford scattering cross section
(for Coulomb forces)

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

NOTATION:

Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega) d\Omega$$

In this case $\sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}$

NOTATION:

Goldstein

Others

$\sigma(\Omega) \leftrightarrow \frac{d\sigma}{d\Omega}$ DIFFERENTIAL SCATTERING CROSS SECTION

$\sigma_T \leftrightarrow \sigma$

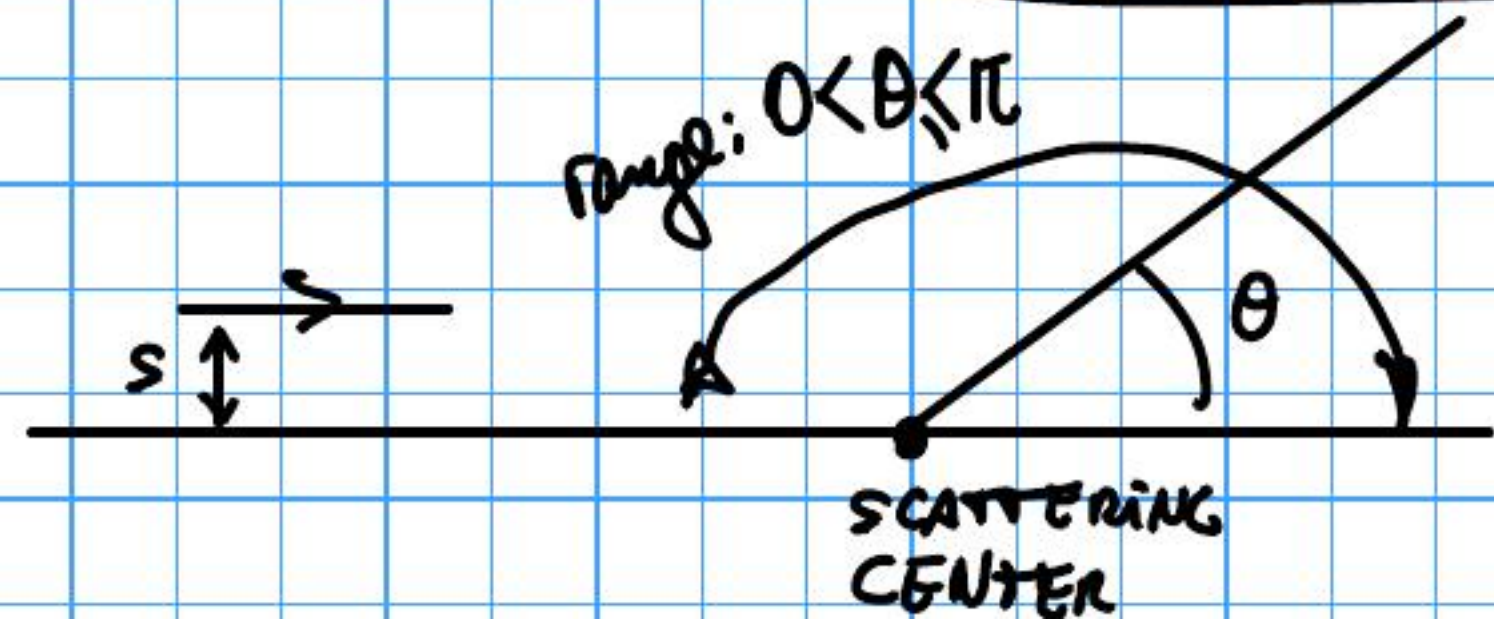
TOTAL SCATTERING CROSS SECTION

$$d\Omega = 2\pi \sin\theta d\theta = 4\pi \sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2} \cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}} d\theta$$

$$\sim \int_0^\pi \frac{d\theta}{(\theta/2)^3}$$

$$= 8 \int_0^\pi \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\pi = +\infty \quad \text{STRONGLY DIVERGENT}$$



CLASSICAL

QUANTUM

$\sigma_T = +\infty$ for any long range potential

$\sigma_T = +\infty$ for some long range potentials
 $V(r)$ do not decay fast enough at $r \rightarrow \infty$

SCATTERING
CENTER

CAVEAT

1) IN A DENSE
CONDUCTING MATERIAL:

$$V = -\frac{k}{r}$$

$$\longrightarrow V^{\text{SCREEN}}(r) = -\frac{k}{r} e^{-r/\lambda}$$

SCREENED COULOMB
POTENTIAL ("YUKAWA")

(In non-conducting materials $V^{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $1/r$)

Chapter 8 Hamiltonian and Hamilton's Eqs of Motion

$$L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \bigg|_{\substack{q_1, \dots, q_n \\ \dot{q}_1, \dots, \dot{q}_n, t}}$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \bigg|_{\substack{q_1, \dots, q_n \\ \dot{q}_1, \dots, \dot{q}_n, t}}$$

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

~~$$L = L(r, \dot{r}, p_\theta)$$~~

~~Lagrange Eqs~~

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

$$f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In Thermodynamics

1st Law

$$dU = dQ + dW =$$

$$T dS - p dV$$

Fluid at constant N

$$U = U(S, V)$$

Internal Energy

We want T, V as variables

We want T, V as variables

$F = A = U - TS$ Helmholtz Free Energy

$$dA = dU - d(TS) = \cancel{TdS} - pdV - \cancel{TdS} - SdT = -SdT - pdV \Rightarrow A = A(T, V)$$

In general

In general

$$f(x,y) = u dx + v dy$$

$$u = \frac{\partial f}{\partial x}$$

$$v = \frac{\partial f}{\partial y}$$

$$(x,y) \longrightarrow (u,y)$$

$$g = f - ux \Rightarrow dg = df - d(ux) = \cancel{u dx} + v dy - \cancel{u dx} - x du = -x du + v dy$$

$$g = t - ux \Rightarrow dg = dt - d(ux) = \cancel{u} dx + v dy - \cancel{u} dx - x du = -x du + v dy$$

$$g = g(u, y)$$

Let's look at $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$

$$p_i = \left. \frac{\partial L}{\partial \dot{q}_i} \right|_{\dot{q}_i = \dot{q}_i}$$

$$H = -\left(L - \sum p_i \dot{q}_i\right)$$

$$dH = dL + \sum d(p_i \dot{q}_i) = -\sum \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i} \right) + \sum (p_i d\dot{q}_i + \dot{q}_i dp_i)$$

$$H = -\left(L - \sum_i p_i \dot{q}_i\right) \quad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \sum_i \left(\dot{q}_i dp_i + p_i d\dot{q}_i \right)$$

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

$$\boxed{\begin{aligned} \frac{\partial H}{\partial p_i} &= \dot{q}_i & \frac{\partial H}{\partial q_i} &= -\frac{\partial L}{\partial q_i} \stackrel{\text{Lagrange Eqs}}{=} -\dot{p}_i & \frac{\partial H}{\partial t} &= -\frac{\partial L}{\partial t} \end{aligned}}$$

CANONICAL EQS OF MOTION